SCC.160: Fundamentals of Communication Systems
Lecture 6

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Outline

- Transmission Modes
- History of Broadcasting
- Analog Audio Broadcasting (AM/FM)
- Digital Audio
- Digital Audio Broadcasting
- Multiple Access in Mobile Networks

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Transmission Modes

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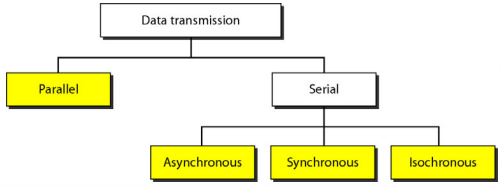
Transmission Modes

- How do we send binary data stream over a wire? Do we send 1 bit at a time; or do we group bits into larger groups?
- The transmission of binary data across a link can be accomplished in either *parallel* or *serial* mode. In parallel mode, multiple bits are sent with each clock tick. In serial mode, 1 bit is sent with each clock tick.
- While there is only one way to send parallel data, there are three subclasses of serial transmission: asynchronous, synchronous, and isochronous.



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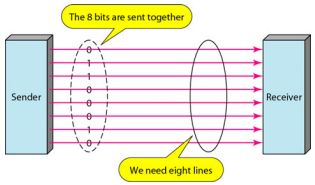
Parallel Transmission

- Binary data, consisting of 1s and 0s, may be organized into groups of *n* bits each.
- Computers produce and consume data in groups of bits. By grouping, we can send data *n* bits at a time instead of 1. This is called *parallel transmission*.
- The mechanism for parallel transmission is a conceptually simple one: use n wires to send n bits at one time. That way each bit has its own wire, and all n bits of one group can be transmitted with each clock tick from one device to another.
- Advantage: speed
- Disadvantage: cost

Figure here shows how parallel transmission works for n=8. Typically, the eight wires are bundled in a cable with a connector at each end.



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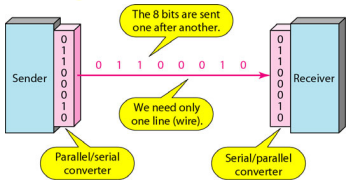
Serial Transmission

- In **serial transmission** one bit follows another, so we need only one communication channel rather than *n* to transmit data between two communicating devices.
- Since communication within devices is parallel, conversion devices are required at the interface between the sender and the line (parallel-to-serial) and between the line and the receiver (serial-to-parallel).
- Advantage: serial transmission reduces the cost of transmission over parallel by roughly a factor of n

Serial transmission occurs in one of three ways: asynchronous, synchronous, and isochronous.



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n=8 Bandwidth = 4000 Bit rate

500

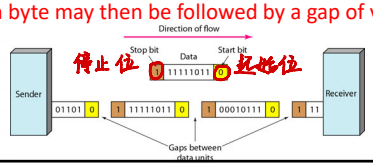
Asynchronous Transmission

异步

起始位=0
停止位=1

- Asynchronous here means "asynchronous at the byte level," but the receiver must still be synchronized with the incoming bit stream.
- Each group, usually 8 bits, is sent along the link as a unit. Without synchronization, the receiver cannot use timing to predict when the next byte will arrive.
- Therefore, in asynchronous transmission, we send 1 start bit (0) at the beginning to alert the receiver to the arrival of a new group. To let the receiver know that the byte is finished, 1 or more stop bits (1s) are appended at the end of each byte.
- In addition, the transmission of each byte may then be followed by a gap of varying duration.

8 bits information + 2 or more extra bits



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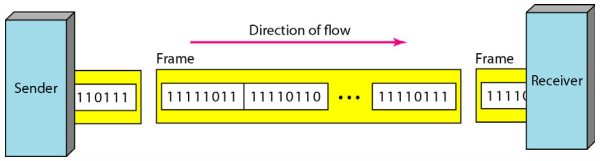
Synchronous Transmission

同步

- In synchronous transmission, the bit stream is combined into longer "frames," which may contain multiple bytes.
- Each byte, however, is introduced onto the transmission link without a gap between it and the next one. It is left to the receiver to separate the bit stream into bytes for decoding purposes.
- There are no start or stop bits.



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异步 = slow, cheap, 低速通信
同步 = fast

Asynchronous vs Synchronous

- The addition of stop and start bits and the insertion of gaps into the bit stream make asynchronous transmission slower than synchronous transmission. But it is cheap and effective - two advantages that make it an attractive choice for situations such as low-speed communication. For example, the connection of a keyboard to a computer is a natural application for asynchronous transmission.
- The advantage of synchronous transmission is speed. With no extra bits or gaps to introduce at the sending end, synchronous transmission is faster than asynchronous transmission. For this reason, it is more useful for high-speed applications such as the transmission of information from one device to another.

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Isochronous Transmission

等时

- Isochronous transmission refers to a method of data transmission where data is sent at a constant and fixed rate. In isochronous communication, there is a continuous and predictable flow of data with a consistent timing pattern. This is particularly important in applications where timing and synchronization are critical, such as in real-time audio and video streaming.
- One of the key characteristics of isochronous transmission is that it prioritizes the timing of data delivery over the integrity of the data itself.
- Isochronous transmission is commonly used in multimedia applications where a steady and uninterrupted flow of data is necessary to maintain the quality of the output.

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History of Broadcasting

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Broadcasting

- What is broadcasting?
- It was defined as "*transmissions intended for reception by the general public, either direct or relayed*" (Source: The U.S. Code of Federal Regulations, title 47, part 97) .
- Private or one-to-one telecommunications transmissions do not qualify under this definition.
- Radio broadcasting
- Radio broadcasting is transmission of **audio** (sound) by *radio waves* intended to reach a wide audience. In terrestrial radio broadcasting, the radio waves are broadcast by a land-based radio station, while in satellite radio, the radio waves are broadcast by a satellite in Earth orbit. To receive the content, the listener must have a broadcast radio receiver (radio).

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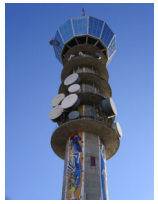
Aab < AM, FM

Broadcasting

Dab < Dab, HD radio, DRM

- Radio broadcasting can be broadly classified into two categories: *analogue audio broadcasting* and *digital audio broadcasting*.
- Analog audio broadcasting includes *AM (amplitude modulation) and FM (frequency modulation)*. [traditional approach]
- Digital audio broadcasting has several different standards: DAB (digital audio broadcasting), HD radio, DRM (Digital Radio Mondiale). [new approach]
- Television broadcasting is a separate service which also uses radio frequencies to broadcast television (video) signals.



Broadcasting tower in Trondheim Norway, By Alex Brasetvik

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Telegraph and Telephone

- Up to the late 19th Century, all the rapid long-distance communications depended upon the telegraph.
- In the 1870s, a rival technology was beginning its development that again changed the face of communication: the *telephone*
- The telegraph and telephone were/are both *wire-based* electrical systems.

Telegraph

Telephone

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The Wireless Revolution

- Long-distance cable communications was expensive and caused signalling problems.
- In 1901, Marconi built a powerful *wireless station* at Cornwall (UK), in preparation for a transatlantic test.
- Many attempts were made to build suitable antennas some of which were destroyed by storms.

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The Wireless Revolution

- On December 12th, 1901, at Signal Hill in St John's (now part of Canada), Marconi received the first *transatlantic radio signal* transmitted from Cornwall, England, a city located about 2000 miles away!
- This was called *radiotelegraphy* or wireless telegraphy. In radiotelegraphy, information is transmitted by pulses of radio waves of two different lengths called "dots" and "dashes", which spell out text messages, usually in Morse code.

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The Wireless Revolution

- In a manual radiotelegraphy system, the sending operator manipulates a switch called a telegraph key which turns the transmitter on and off, producing the pulses of radio waves. At the receiver, the pulses are audible in the receiver's speaker as beeps, which are translated back to text by an operator who knows Morse code.

This type of radio was used exclusively for long-distance person-to-person text communication for commercial, diplomatic and military purposes and hobbyists; broadcasting did not exist. Audio was not carried.

By the U.S. Army Signal Corps - Wikipedia

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Audio Broadcasting

- For *audio broadcasting* to be possible, electronic detection and amplification devices had to be incorporated.
- In late 1906, Reginald Fessenden made the first radio broadcast of entertainment and music, although a lack of verifiable details has led to some doubts about this claim.
- In The Hague, the Netherlands, PCGG (a radio station) started broadcasting on November 6, 1919, making it, arguably the first commercial broadcasting station.

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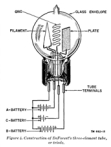
Audio Broadcasting

- Broadcasting began with AM radio, which came into popular use around 1920 with the spread of vacuum tube radio transmitters and receivers. Before this, all forms of electronic communication (early radio, telephone, and telegraph) were *one-to-one*, with the message intended for a single recipient.
- By the early 1920s radio broadcasting became a household medium, at first on the AM band and later on FM.



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The Golden Age of Radio

- AM radio remained the dominant method of broadcasting for the next 30 years, a period called the "Golden Age of Radio", until television broadcasting became widespread in the 1950s and received most of the programming previously carried by radio.

According to a 1947 C. E. Hooper survey, 82 out of 100 Americans were found to be radio listeners.

By Franklin D. Roosevelt Library Public Domain Photographs - Wikipedia

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Analog Broadcasting

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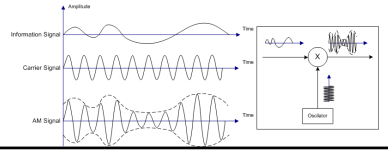
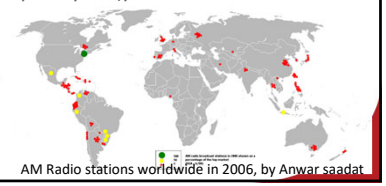
AM Broadcasting

- AM stations were the earliest broadcasting stations to be developed.
- AM refers to *amplitude modulation*, a mode of broadcasting radio waves by varying the amplitude of the carrier signal in response to the changing amplitude of the modulating signal.
- If the modulating signal is a single-frequency sine wave, the modulated signal has three components: *the carrier wave c(t) with frequency fc, and two sidebands* with frequencies slightly above and below the carrier frequency fc. (you should have noticed this in the lab simulations)





AM Radio stations worldwide in 2006, by Anwar saadat

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AM Transmitter Block Diagram

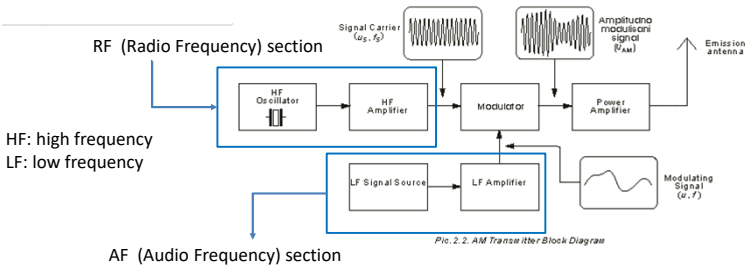

Fig. 2.2. AM Transmitter Block Diagram

- AF section: LF modulating signal comes from microphone or other signal resources, amplified in the LF amplifier.
- RF section: HF oscillator creates HF carrier wave which gets amplified in the HF amplifier to the required signal level.
- Modulator: the high-frequency carrier and low-frequency modulating signal feed into modulator and the output signal is the AM signal, which will be amplified in the power amplifier and then led to the antenna.

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Frequency

载波频率：由发射台设计

- The frequency of the transmitting station, specifically the carrier frequency, is a fixed value that is determined by the station's design and regulatory standards. In the context of amplitude modulation (AM) broadcasting, the carrier frequency is typically assigned by regulatory bodies and falls within a specific frequency band.
- For example, in the medium-frequency (MF) band used for standard AM broadcasting, the carrier frequencies are assigned within the range of 530 to 1700 kHz in the United States. The exact carrier frequency for a particular station is determined by the station's license and the available frequencies within the allocated band.
- In the frequency equation for AM modulation mentioned earlier:
- $f_{transmitting}=f_c$
- Here, $f_{transmitting}$ represents the carrier frequency of the transmitting station.

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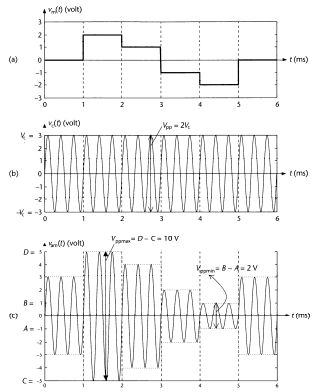

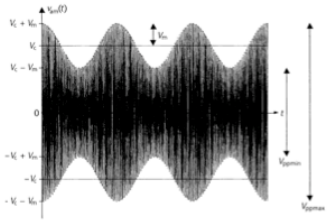

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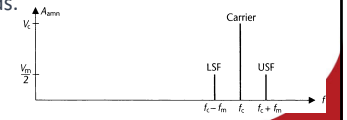
Transmission

Upper sideband & Lower sideband

- In amplitude modulation (AM) for radio broadcasting, two sidebands are generated on either side of the carrier frequency. These sidebands are referred to as the upper sideband (USB) and the lower sideband (LSB). Both sidebands contain the same information, but they are mirror images of each other.
- The standard AM broadcast signal consists of the carrier frequency and both sidebands. However, to conserve bandwidth and power, only one sideband is transmitted, along with the carrier. The selection of the sideband to be transmitted is usually based on practical considerations and standards.



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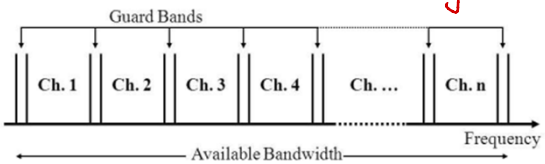
Standard Bandwidth Allocation for AM Radio

- AM stations are allowed carrier frequencies anywhere between 530 and 1700 kHz (1.7 MHz). However, each station's carrier frequency must be separated from those on either side of it by at least 10 kHz (**guard band**) to avoid interference.
- For example, if one station uses a carrier frequency of 1100 kHz, the next station's carrier frequency cannot be lower than 1110 kHz.

guard band



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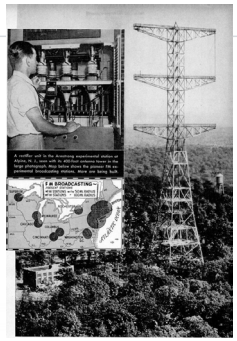
FM > AM, FM can avoid amplitude-based interface.

The Dawn Of FM Radio: Edwin Armstrong

Even though AM radio improved significantly, it still had major limitations:

- static interference from household appliances and lighting
- limited audio quality (non-ideal power amplifiers with small dynamic range)
- nighttime interference between many stations (co-channel interference), because of ionospheric refraction

FM broadcasting is a method of radio broadcasting using frequency modulation (FM), invented in 1933 by American engineer Edwin Armstrong, which eliminates the above problems.



One of the first FM radio stations, Edwin Armstrong's experimental station W2XMN in Alpine, New Jersey, USA.--- By E. W. Murtfeldt

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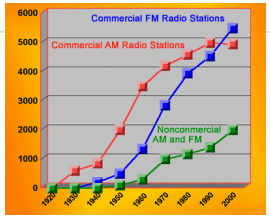
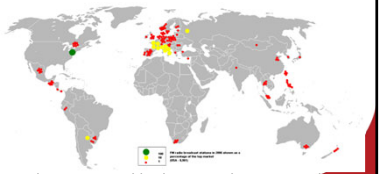
FM Broadcasting

- By 1941, 50 FM stations were on the air. FM radio didn't really take off until after Armstrong's death in 1954.
- But after 2000, FM becomes more popular than AM radio.
- Due to its superior quality, FM radio transmission is still one of the most popular transmission methods.
- Its price is so cheap it can be integrated into many consumer electronic devices, e.g., iPod nano, most android mobile phones



http://www.cybercollege.com/frtv/frtv020.htm



FM Radio stations worldwide in 2006, by Anwar saadat

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Digital Audio

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Analog Audio vs Digital Audio

- AM and FM broadcastings are **analogue** audio technologies.
- An analogue audio system converts physical waveforms of sound into electrical representations of those waveforms by use of a transducer, such as a microphone. The sounds are then *stored on an analogue medium* such as magnetic tape, or *transmitted through an analog medium* such as a telephone line or radio. The process is reversed for reproduction: the electrical audio signal is amplified and then converted back into physical waveforms via a loudspeaker.
- Analogue audio retains its fundamental *wave-like characteristics* throughout its storage, transformation, duplication, and amplification.

magnetic tapes

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Analog Audio vs Digital Audio

- *Analog audio signals are susceptible to noise and distortion*, due to the innate characteristics of electronic circuits and associated devices. Disturbances in a digital system do not result in error unless the disturbance is so large as to result in a symbol being misinterpreted as another symbol or disturb the sequence of symbols. It is therefore generally possible to have an entirely *error-free digital audio system* in which no noise or distortion is introduced between conversion to digital format and conversion back to analogue.
- Also, digital audio signal may optionally be encoded for correction of any errors that might occur in the storage or transmission of the signal. This technique, known as *channel coding*, is essential for broadcast or recorded digital systems to maintain bit accuracy. Eight-to-fourteen modulation is a channel code used in the audio compact disc (CD). (https://www.physics.udel.edu/~watson/scen103/efm.html)

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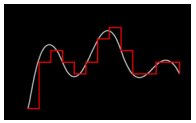
Digital Audio

- Digital audio is *a representation of sound recorded in, or converted into, digital form*. In digital audio, the sound wave of the audio signal is typically encoded as numerical samples in a continuous sequence. For example, in CD audio, samples are taken 44,100 times per second, each with 16-bit resolution.

- Following significant advances in digital audio technology during the 1970s and 1980s, it gradually replaced analogue audio technology in many areas of audio engineering and telecommunications in the 1990s and 2000s.



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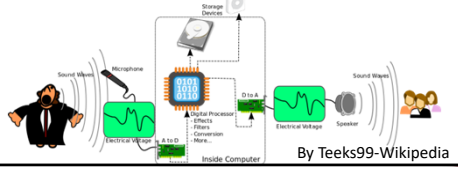
Digital Audio

- In a digital audio system, an analogue signal representing the sound is converted with an analogue-to-digital converter (ADC) into a digital signal, typically using pulse-code modulation (PCM).
- A digital audio signal may be stored or transmitted. Digital audio can be stored on a CD, a digital audio player, a USB flash drive, or any other digital data storage device.
- For playback, digital audio must be converted back to an analogue signal with a DAC. According to the Nyquist sampling theorem, with some practical and theoretical restrictions, the original analogue signal can be accurately reconstructed from the digital signal.



By Teeks99-Wikipedia

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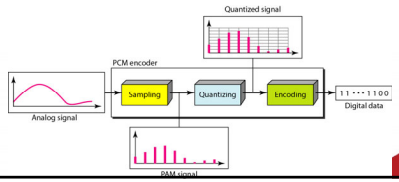
Recap: PCM

- The conversion of an analogue data into a discrete-valued, discrete-time (digital signal) is accomplished in 3 steps
  - Sampling: the analog signal is sampled every T_s seconds, where T_s is the sample interval or period. (the sampling rate must obey the Nyquist theorem)
  - Quantization and encoding: where the sampled values are quantized and then converted to one of n_b-bit code words.
- The above digitization is generally known as Pulse Code Modulation (PCM) (more details have been introduced in Lecture 5).



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CD Sample Rate

- Q: CD samples the left and right channel audio waveforms each at 44.1KHz (44.1K samples per second) to take into account real filter roll-off characteristics. Quantisation is 16 bits linear. Calculate the bit rate.

- Since Bit rate = sampling rate * number of bits per sample = $f_s \times n_b$,
- Hence CD bit-rate = 2 * 44100 * 16 = 1.41 Mb/s

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Digital Audio Broadcasting

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Digital Audio Broadcasting

- Digital audio can be used in *audio broadcasting*. Standard technologies include Digital audio broadcasting (DAB), Digital Radio Mondiale (DRM), HD Radio and In-band on-channel (IBOC).
- DAB is a digital radio standard for broadcasting digital radio services in many countries around the world.
- DAB is more robust with regard to noise and multipath fading for mobile listening, compared to analogue broadcastings. But DAB reception quality degrades rapidly when the signal strength falls below a critical threshold, whereas FM reception quality degrades slowly with the decreasing signal, providing effective coverage over a larger area.

By Matt Kieffer

By Håvard Wien

By ROBERTS

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Digital Audio Broadcasting

- DAB can give substantially *higher spectral efficiency* than analogue systems. In many places, this has led to an increase in the number of stations available to listeners, especially outside of the major urban areas.
- Existing FM transmissions
  - Each transmitter has a bandwidth of 0.2 MHz
  - Nearby transmitters must be 0.4 MHz apart
  - 2.2 MHz needed for a network covering entire country
- DAB
  - 1.5 MHz for 10 services covering entire country using a single frequency network
  - *15 times* more efficient !

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Multiple Access in Mobile Networks

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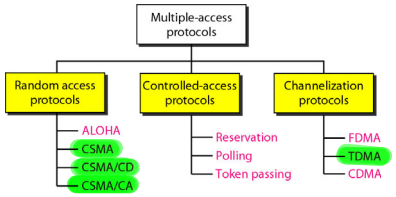
Multiple Access Protocols 多接入协议

- When nodes or stations are connected and use a common link, called a broadcast link, we need a multiple-access protocol to coordinate access to the link.
- Many protocols have been devised to handle access to a shared link.
- We categorize them into three groups



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Channel Partitioning Protocols

- Channelization protocols (or channel partitioning, as it is sometimes called) are multiple-access methods in which the available bandwidth of a link is shared in time, frequency, or through code, among different stations/devices.
- We will discuss three channelization protocols: FDMA, TDMA, and CDMA.

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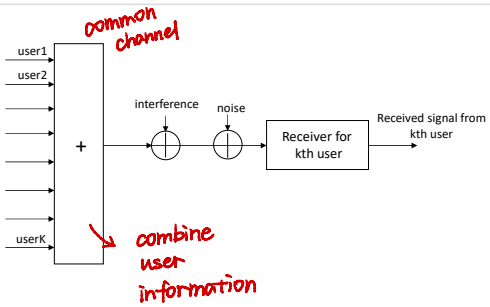
Generic Multiple Access Communication System



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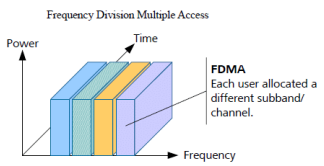
用户用不同的频段实现高效率通信

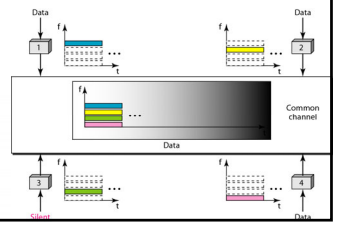
Frequency Division Multiple Access (FDMA)

频分多址

- In frequency-division multiple access (FDMA), the available bandwidth is divided into frequency bands. Each station/device is allocated a band to send its data. In other words, *each frequency band is reserved for a specific station, and it belongs to the station all the time.*
- To prevent station interferences, the allocated bands are separated from one another by small guard bands.



In FDMA, the available bandwidth of the common channel is divided into bands that are separated by guard bands.



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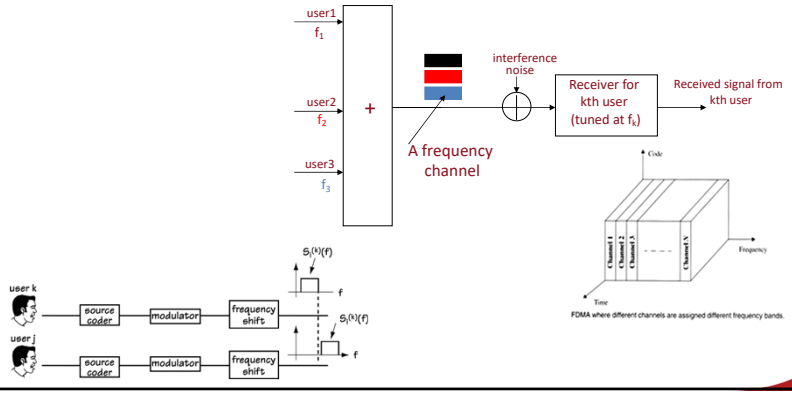
Frequency Division Multiple Access (FDMA)



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FDMA Application

- FDMA allows multiple users to send data through a single communication channel, such as a coaxial cable or microwave beam, by dividing the bandwidth of the channel into separate non-overlapping frequency sub-channels/bands and allocating each sub-channel/band to a separate user. (source: wikipedia)
- Users can send data through a subchannel by modulating it on a carrier wave at the subchannel's frequency. FDMA is used in satellite communication systems and cellular technologies like 2G GSM.

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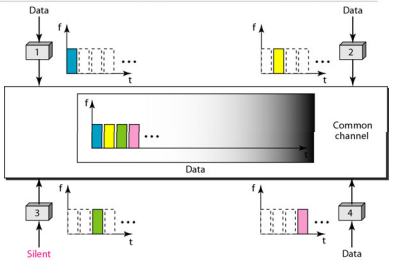
用户在不同的时间段上进行交替

Time Division Multiple Access (TDMA)

时分多址

- In time-division multiple access (TDMA), the stations/devices share the bandwidth of the channel in time. Each station/device is allocated a time slot during which it can send data.
- Each station transmits its data in its assigned time slot.
- The main problem with TDMA lies in achieving synchronization between the different stations. Each station needs to know the beginning of its slot and the location of its slot.



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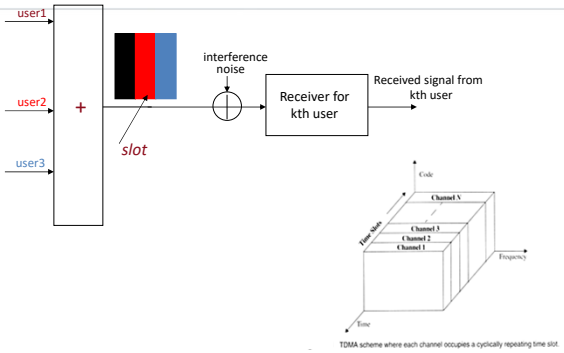
Time Division Multiple Access (TDMA)



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每个用户用不同的伪随机码对数据进行编码

TDMA Application

- TDMA allows several users to share the same frequency channel by occupying different time slots.
- The users transmit in rapid succession, one after the other, each using its own time slot. This allows multiple stations to share the same transmission medium (e.g. radio frequency channel) while using only a part of its channel capacity.
- TDMA is used in the 2G cellular systems such as GSM. TDMA was first used in satellite communication systems by Western Union in its Westar 3 communications satellite in 1979. It is now used extensively in satellite communications.

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Code Division Multiple Access (CDMA)

码分多址

只有一个channel，且所有station可同时发送数据

- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link. It differs from TDMA in that all stations can send data simultaneously; there is no timesharing.
- CDMA simply means communication with different codes.
- *Real-life analogy:* In a large room with many people, two people can talk privately in English if nobody else understands English. Another two people can talk in Chinese if they are the only ones who understand Chinese, and so on. In other words, the common channel, i.e., the space of the room in this case, can easily allow communication between several couples, but in different languages (codes).

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Code Division Multiple Access (CDMA)

- **Idea of CDMA:**
- Let us assume we have four stations, 1, 2, 3, and 4, connected to the same channel. The data from station 1 are d1, from station 2 are d2, and so on. The code assigned to the first station is c1, to the second is c2, and so on.
- We assume that the assigned codes have two properties:
  1. If we multiply each code by another, we get 0. $\boxed{c_i \cdot c_j = 0}$
  2. If we multiply each code by itself, we get 4 (the number of stations). $\boxed{c_i \cdot c_i = 4}$
- Then, let us see how these four stations send data using the same common channel.

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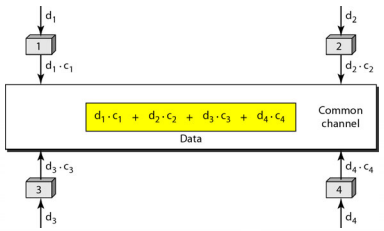
Code Division Multiple Access (CDMA)

- **Idea of CDMA (cont'd):**
- Station 1 multiplies (a special kind of multiplication, as we will see) its data by its code to get d1 · c1. Station 2 multiplies its data by its code to get d2 · c2, and so on. The data that go on the channel are the sum of all these terms, as shown in the box.



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Code Division Multiple Access (CDMA)

Lancaster University

- **Idea of CDMA (cont'd):**
- Any station that wants to receive data multiplies the data on the channel by the code of the sender.
- For example, suppose stations 1 and 2 are talking to each other. Station 2 wants to hear what station 1 is saying. It multiplies the data on the channel by c1, the code of station 1. Because (c1 · c1) is 4, but (c2 · c1), (c3 · c1), and (c4 · c1) are all 0s, station 2 divides the result by 4 to get the data from station 1.

$$\text{data} = (d_1 \cdot c_1 + d_2 \cdot c_2 + d_3 \cdot c_3 + d_4 \cdot c_4) \cdot c_1$$
$$= d_1 \cdot \underbrace{c_1 \cdot c_1}_{4} + d_2 \cdot \underbrace{c_2 \cdot c_1}_{0} + d_3 \cdot \underbrace{c_3 \cdot c_1}_{0} + d_4 \cdot \underbrace{c_4 \cdot c_1}_{0} = 4 \times d_1$$

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Code Division Multiple Access (CDMA) (example)

Lancaster University

- In CDMA, each station is assigned a unique code, which is a sequence of numbers called chips. Some example chips are shown below.

C_1	C_2	C_3	C_4
[+1 +1 +1 +1]	[+1 -1 +1 -1]	[+1 +1 -1 - 1]	[+1 -1 -1 +1]

- These codes are carefully chosen. They are called *orthogonal sequences* and have the following properties:
  - Each sequence is made of N elements, where N is the number of stations.
  - If we multiply a sequence by a number, every element in the sequence is multiplied by that number. For example, 2 • [+1 +1 −1 −1] = [+2 +2 −2 −2]
  - If we multiply two equal sequences, element by element, and add the results, we get N. In other words, the inner product of two equal sequences is N. For example, [+1 +1 −1 −1] • [+1 +1 −1 −1] = 1 + 1 + 1 + 1 = 4
  - If we multiply two different sequences, element by element, and add the results, we get 0. The inner product of two different sequences is 0. This is due to the orthogonality.

自身×自身=4
自身×其它=0

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Code Division Multiple Access (CDMA)

Lancaster University

- Data Representation Rules:
- We follow these rules for encoding: If a station needs to send a 0 bit, it encodes it as −1; if it needs to send a 1 bit, it encodes it as +1. When a station is idle, it sends no signal, which is interpreted as a 0.

Data bit 0 → -1	Data bit 1 → +1	Silence → 0

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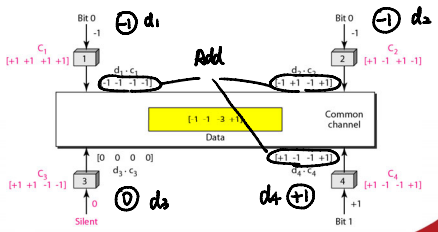
Code Division Multiple Access (CDMA) - Example

Lancaster University

- We assume that stations 1 and 2 are sending a 0 bit and station 4 is sending a 1 bit. Station 3 is silent. Correspondingly, they are translated to −1, −1, 0, and +1.
- Each station multiplies the corresponding number by its chip (its orthogonal sequence), where the new sequence is sent to the channel. Assume that all stations send the resulting sequences at the same time. The sequence on the channel is the sum of all four sequences.

Suppose that station 3 is listening to station 2. Station 3 multiplies the total data on the channel by the code for station 2, which is [+1 −1 +1 −1]. It gets [−1 −1 −3 +1] • [+1 −1 +1 −1] = −4/4 = −1 → bit 0



3号 = 总数据 × 2号站

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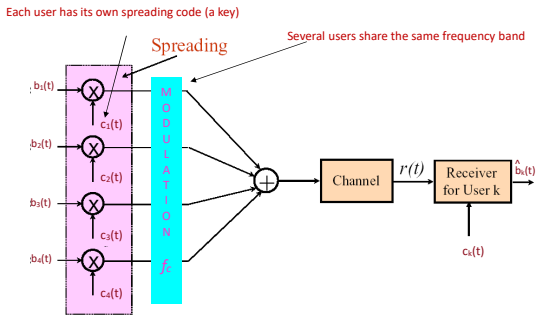
Code Division Multiple Access (CDMA)



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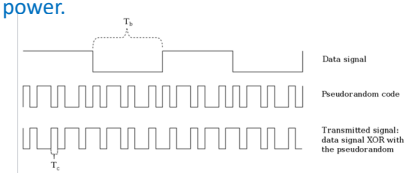
CDMA Application

- CDMA is a spread-spectrum multiple-access technique. It spreads the bandwidth of the data uniformly for the same transmitted power.
  - The original data signal has a pulse duration of $T_b \Longrightarrow$ the bandwidth is $1/T_b$
  - The transmitted signal after multiplying with code has a pulse duration of $T_c \Longrightarrow$ the bandwidth is $1/T_c$
  - Since $T_c < T_b$, the bandwidth of the transmitted signal is much larger than the original signal.



The ratio T_c and T_b is called the spreading factor and determines to a certain extent the upper limit of the total number of users supported simultaneously

- CDMA is used as the access method in many mobile phone standards. IS-95, also called "cdmaOne", and its 3G evolution CDMA2000 use CDMA as their radio technologies.

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Thank You!

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